Dimensionality Reduction as Alpha

Linear vs. Deep Feature Extractors for Tree-Based Cross-Sectional Return Prediction

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Bachelor Thesis Defence

in collaboration with  Thira Labs

Motivation & Research Questions

Motivation & the conjecture

ML reshaped asset pricing — Gu, Kelly & Xiu (2020): nonlinear ML — tree-based methods and neural networks — gives economically significant OOS forecasts (long–short Sharpe $\approx$1.35, $\sim$2$\times$ a leading regression).

Latent-factor hypothesis Ross 1976

$$r_{i,t} = \boldsymbol{\beta}_i^\top \mathbf{f}_t + \epsilon_{i,t}, \quad K \ll N$$

80 alpha signals each proxy a few common factors $\Rightarrow$ highly redundant.

So, in theory: compress first $\rightarrow$ drop noise, keep signal $\rightarrow$ $\uparrow$Rank IC, $\uparrow$Sharpe.

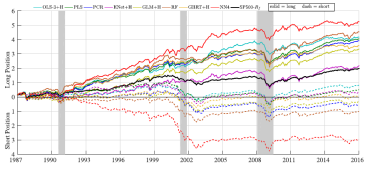


Figure 9
Cumulative return of machine learning portfolios
The figure shows the cumulative log returns of portfolios sorted on out-of-sample machine learning return forecasts. The solid and dashed lines represent long (top decile) and short (bottom decile) positions, respectively. The shaded periods show NBER recession dates. All portfolios are value weighted.

Source: GKX (2020), *RFS* — cumulative return of ML-sorted portfolios.

Downloaded from https://academ

The question — and the gap

Central question

Should a high-dimensional alpha library be compressed before prediction, and by which method — linear (PCA/FA) or deep (GKX-AE / FactorVAE)?

Research questions

RQ1 DR vs. the raw-alpha baseline?

RQ2 Deep vs. linear — does complexity pay?

RQ3 Normalisation × universe breadth?

RQ4 Do IC gains become Sharpe gains?

The gap

- 1. Deep DR tested only as *end-to-end* models.
- 2. No controlled 5-way comparison.
- 3. Prior tests leak (refit on full sample).

This thesis: extractor role · 5-way · strict walk-forward

A quick finance primer

The scorecard — the Sharpe ratio



Return alone is misleading — a strategy that earns well but swings wildly can wipe you out.

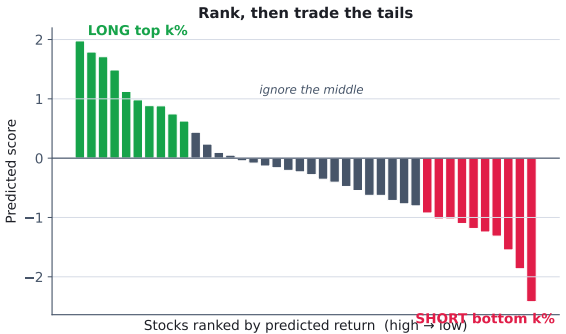
The Sharpe ratio rescales return by the risk taken to earn it:

$$\text{Sharpe} = \frac{\mathbb{E}[R] - R_f}{\sigma_R}\sqrt{252}$$

- Same average return $\Rightarrow$ smoother book wins.
- Annualised; higher is better. Textbook daily convention is $\sqrt{252}$; the backtester annualises over *calendar* days, so all reported figures carry $\sqrt{365}$.
- Rule of thumb: > 1 good, > 2 excellent.

This thesis judges everything on OOS Sharpe.

Turning predictions into a portfolio — long–short



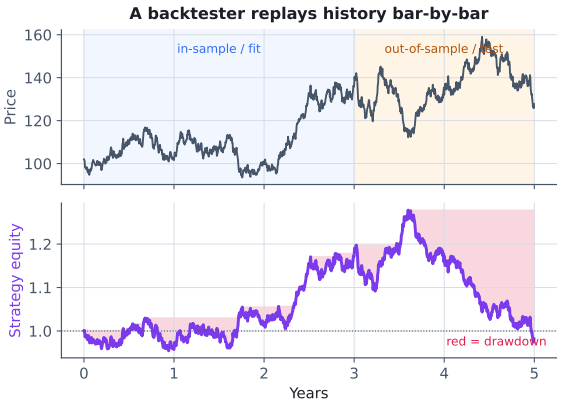
The model outputs a predicted return for every stock each day. We don't trade the number — we trade the *ranking*:

- Buy (long) the **top** $k\%$ — expected winners.
- Sell (short) the **bottom** $k\%$ — expected losers.
- Ignore the ambiguous middle.

Long and short legs are sized to cancel ⇒ a dollar-neutral book that profits from the *spread*, not the market's direction.

Here $k = 25\%$ for clarity; the thesis uses $k = 5\%$.

Testing the idea on history — the backtester



A backtester replays the past *bar by bar*: at each day it feeds the model only past data, forms the long–short book, and books the next day's P&L — after trading costs.

- In-sample: fit the model.
- Out-of-sample: judge it on *unseen* data.
- Drawdowns show the pain along the way.

Takeaway. No look-ahead: every decision uses only information available *at that moment* — the guardrail against fooling yourself.

Methodology

Data & asset universes

ETF universe

60 ETFs — sectors, regions, fixed income, commodities.
Daily 2006–2025 · OOS **2019–2025** (7 folds) · ∼105k obs.

S&P 500 universe

483 large caps — post-pandemic regime.
Daily 2020–2025 · OOS **2022–2025** (4 folds) · ∼484k obs.

80 alpha signals

What's an alpha? A short formula over price/volume data that scores each stock ⇒ a raw predictive feature, not a finished strategy.

e.g. Alpha101: $\frac{\text{close} - \text{open}}{\text{high} - \text{low} + 0.001}$ — where in the day's range did it close? (intraday momentum)

Curated from WorldQuant **101 Formulaic Alphas** Kakushadze 2016, per Jansen (2020) — price/volume only ⇒ comparable across universes.

Target: forward daily return $\text{close}_{t+1} \to \text{close}_{t+2}$, market-on-close.

Source: Yahoo Finance via `yfinance`

Five feature configurations

Config	Family	What it does
Baseline	none	80 standardised raw alphas — benchmark to beat
PCA	linear	orthogonal max-variance components
FA	linear	shared variance vs. idiosyncratic noise
GKX-AE	deep	deterministic nonlinear factors, linear bottleneck
FactorVAE	deep	probabilistic, KL-regularised latent factors

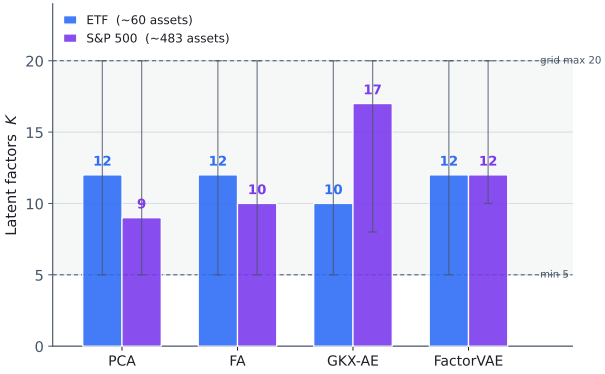
Track A — Baseline/PCA/FA

tuned *jointly* with the tree.

Track B — GKX-AE/FactorVAE

encoder tuned first, latent *frozen*, then fed to the tree.

How much compression? — Latent factors selected



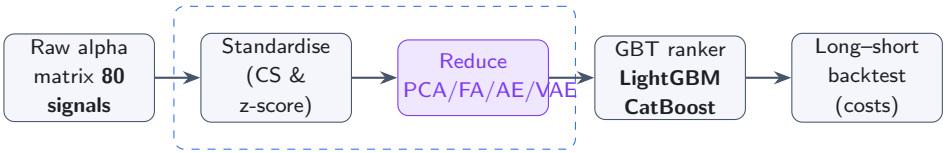
Bars = median K per fold · whiskers = full tuning range.

K is tuned, not fixed — Optuna re-picks it every fold on validation Sharpe (grid 5–20).

- Wide S&P 500 $\Rightarrow$ deep models add factors (GKX-AE ≈ 17).
- Linear stays lean (PCA ≈ 9): variance saturates.
- FactorVAE steady (≈ 12): KL prior caps factors.

Takeaway. No universal K — the right amount of compression tracks universe breadth and extractor type, not theory.

Pipeline & leakage-free evaluation



Strict walk-forward

Annual OOS folds, capped 4-yr train. Every transform fit only on train, applied frozen — zero leakage.

Selected on economics

Optuna tunes on validation Sharpe, not RMSE (uninformative at this signal-to-noise).

The ranker — LightGBM vs. CatBoost

Both are gradient-boosted decision trees used as the cross-sectional ranker — I run both and let validation Sharpe pick.

	LightGBM	CatBoost
Tree growth	leaf-wise (best-first), asymmetric	symmetric / oblivious (balanced)
Boosting	standard gradient boosting	ordered boosting — cuts target leakage
Speed	very fast (histogram, GOSS, EFB)	slower, but robust out-of-the-box
Overfitting	flexible ⇒ needs careful reg.	resistant on small / noisy data
Best when	large data, speed matters	noisy signal, low signal-to-noise

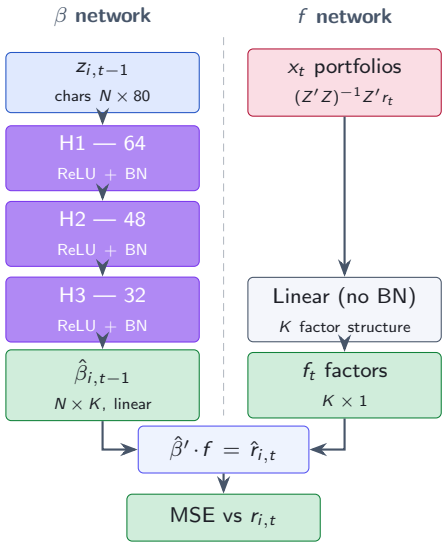
Why two?

LightGBM's leaf-wise growth chases the sharpest split; CatBoost's ordered boosting trades a little flexibility for less overfitting on our low-signal

Same role

Neither sees raw prices — both consume the (reduced) feature block and output a per-stock rank, tuned only on the train fold.

Conditional Autoencoder — My Implementation



✓ **Faithful to GKX (2021):**

- **Input**: firm characteristics $z_{i,t-1}$
- **Decoder**: single linear factor layer
- **Factors**: from managed portfolios x_t

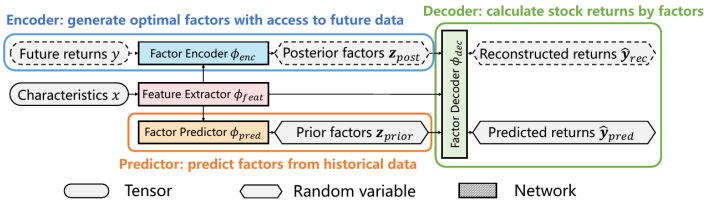
× **My departures:**

- **Early stop** on Rank IC
  not validation MSE
- **No ensembling**
  paper averages 10 seeds

$$r_{i,t} = \hat{\beta}'_{i,t-1} f_t + u_{i,t}$$

Deterministic. No stochastic regulariser — overfit-prone on the narrow cross-section.

Deep extractor II — FactorVAE



Source: Duan et al. (2022), *AAAI* — prior–posterior VAE: an encoder sees future returns to form an "optimal" posterior that guides the prior.

Probabilistic. Encodes a short look-back into a stochastic latent of disentangled factors; trained on the ELBO.

KL regularisation keeps the latent space smooth — the decisive difference from GKX-AE.

Evaluation & statistical testing

Metrics

Sharpe ratio of the dollar-neutral long–short book (top/bottom 5%, after costs) **(primary)**.

Rank IC / ICIR — Spearman corr. of predicted vs. realised ranks (diagnostic).

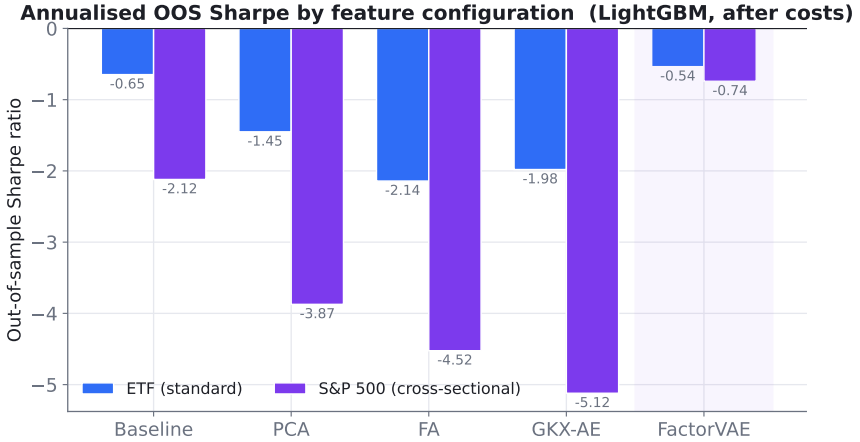
Significance

Block-bootstrap (5,000 resamples, mean block 21 days) tests of Sharpe differences vs. the Baseline; Newey–West HAC on Rank IC and on the market-orthogonality regressions.

Takeaway. Everything is judged on out-of-sample portfolio economics, with serial-correlation-robust significance — not on in-sample fit.

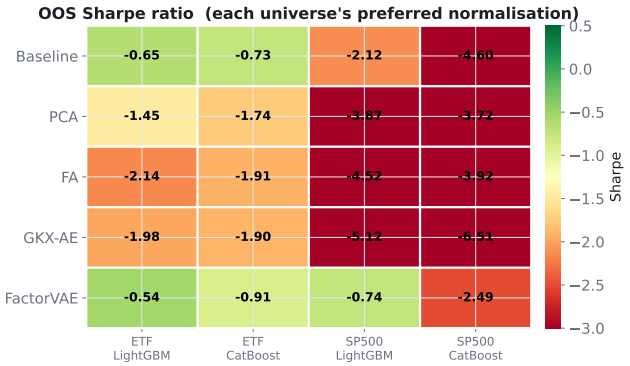
Results

Headline — OOS Sharpe by configuration



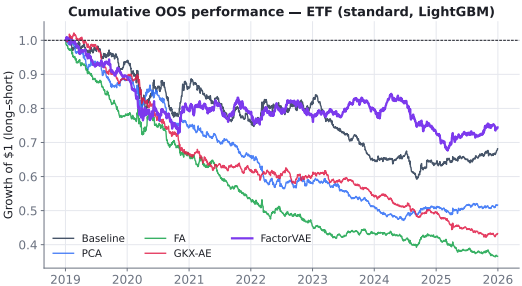
Takeaway. GKX-AE, PCA and FA never beat the **Baseline** *significantly*; FactorVAE is the exception — it edges out Baseline in *both* markets, significantly so on the S&P 500 ($p=0.008$), not on the narrower ETF universe ($p=0.79$).

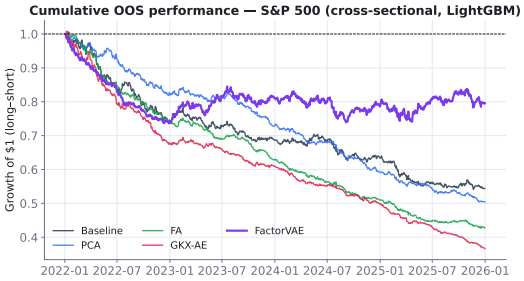
The full grid



- FactorVAE = greenest row (best/tied in 3 of 4 columns).
- GKX-AE & FA = reddest — depth did *not* buy quality.
- Every cell < 0: genuinely unprofitable OOS after costs.

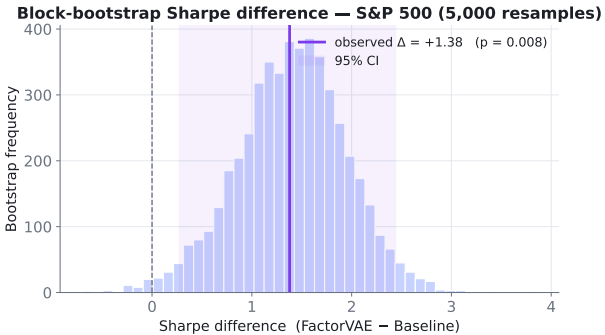
Both universes through time





Takeaway. FactorVAE (thick purple) is the steadiest line in both; the gap is largest on the *wide* S&P 500, where its regularisation pays off most.

Is FactorVAE's edge real? — Block bootstrap

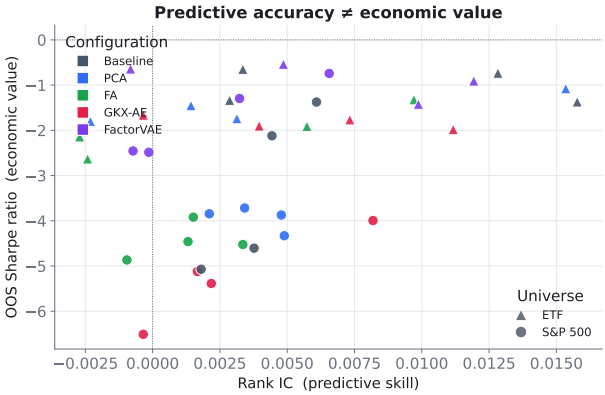


Sharpe diff **FactorVAE** − **Baseline**, 5,000 resamples:

Universe	ΔSharpe	p
S&P 500	+1.38	0.008
ETF	+0.11	0.79

Significant on the S&P 500 ($p < 0.01$, stable for block lengths of 5–63 days); present but not significant on the narrow ETF universe.

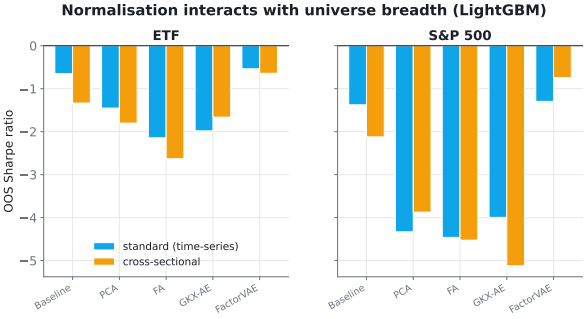
Predictive accuracy ≠ economic value



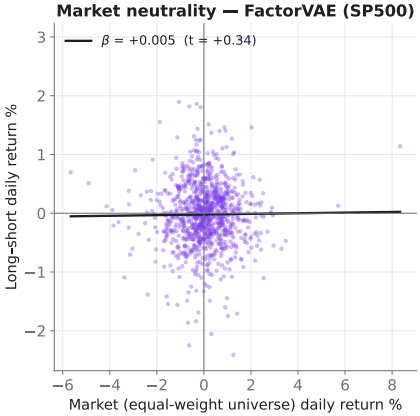
Each point is one configuration. Rank IC and Sharpe barely line up — the highest-IC config is often *not* the best Sharpe.

Implication: validate reducers on portfolio economics, never on a predictive metric alone.

Normalisation matters; losses are market-neutral

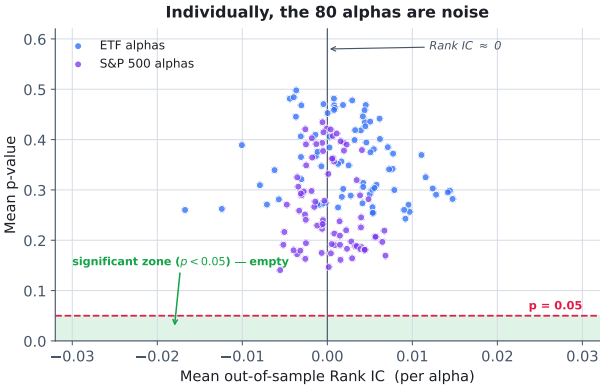


std wins on narrow ETF · on wide S&P 500 it's mixed — **cross-sectional** only helps FactorVAE.



$\beta \approx 0$, tiny R^2 — genuine negative alpha, not market exposure.

Why so weak? — The raw alphas barely predict



Per-alpha OOS statistics

	ETF	S&P 500
Median |Rank IC|	0.005	0.003
Median p-value	0.35	0.25
Significant ($p < 0.05$)	0%	0%

- **No alpha is significant** OOS — ICs near zero, every $p > 0.05$.
- Faint, diffuse signal — only the tree's *combination* adds value, so compressing first discards it.

Takeaway. Near-efficient universe — weak, non-significant inputs explain the negative OOS Sharpes across *all* configurations.

What we learned

F1 — Mostly no free lunch

PCA/FA/GKX-AE never beat **Baseline** significantly; only FactorVAE does — on the S&P 500, under *both* learners.

F2 — Regularisation > depth

FactorVAE leads; GKX-AE trails. The KL regulariser, not depth, is decisive.

F3 — IC $\neq$ Sharpe

Accuracy is a poor proxy for economic value. Always judge on the portfolio.

Bottom line

Feature compression is a *conditional* tool — valuable on wide universes with the right regulariser, harmful otherwise — not a guaranteed gain conferred by theory.

real but conditional

Thank you — questions welcome.

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